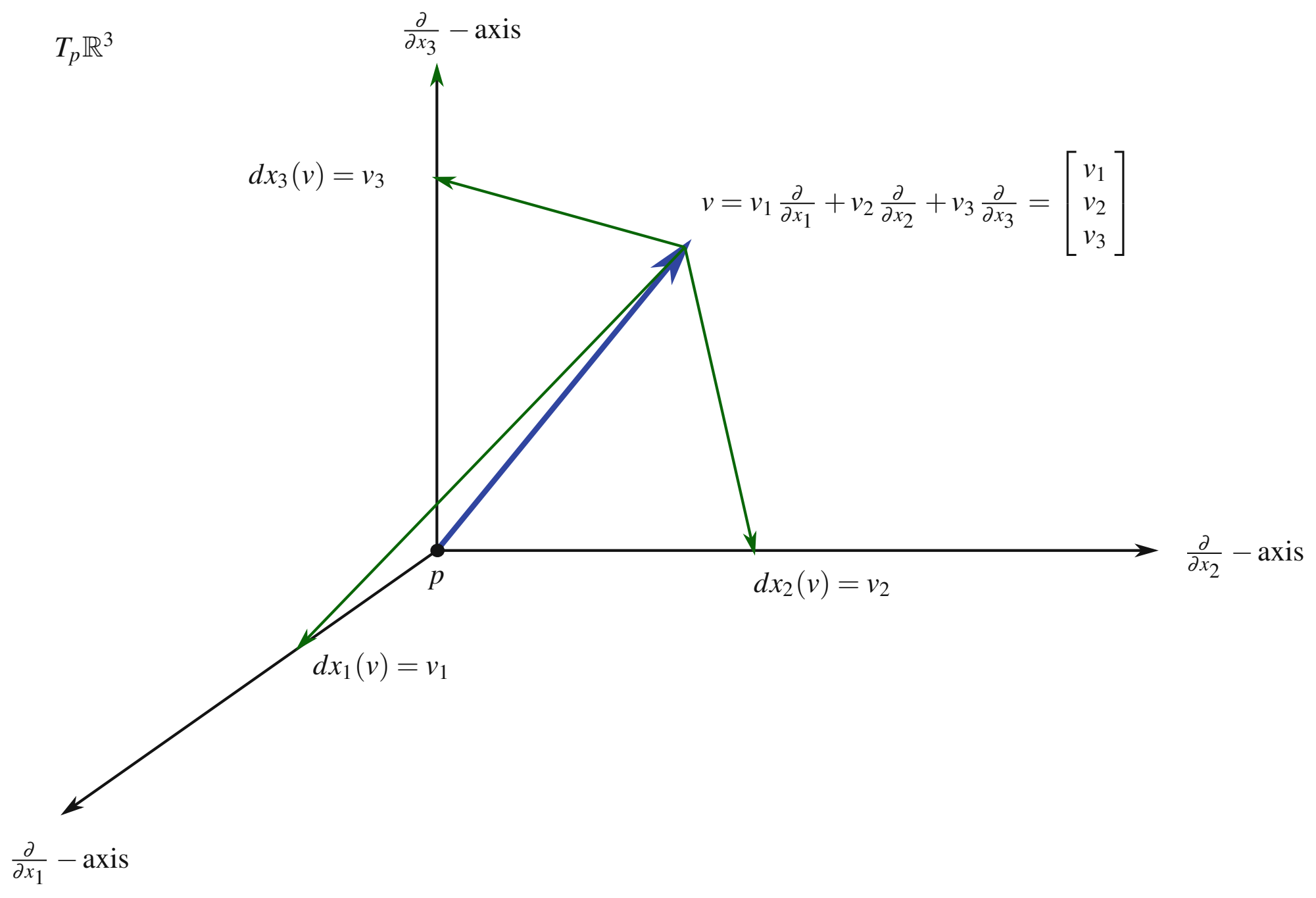

$T_p\mathbb{R}^3$
$\frac{\partial}{\partial x_3}$ – axis
$dx_3(v) = v_3$
$v = v_1 \frac{\partial}{\partial x_1} + v_2 \frac{\partial}{\partial x_2} + v_3 \frac{\partial}{\partial x_3} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$
$\frac{\partial}{\partial x_2}$ – axis
p
$dx_2(v) = v_2$
$dx_1(v) = v_1$
$\frac{\partial}{\partial x_1}$ – axis